

GIOVANNI MARRAFFINI

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RESEARCH EXPERIENCE

PhD Candidate | INRIA

March 2026 - Present

Foundation models pretraining and evaluation for understanding cognition

Paris, France

First comprehensive benchmarking of fMRI foundation models, evaluating via probing whether internal representations retain cognitively relevant variance. Followed by targeted fine-tuning to recover representation quality, identifying a core reconstruction-vs-representation trade-off.

Research Engineer | Paris Brain Institute

May 2025 - March 2026

Applied EEG time-series foundation models for neurodegenerative diseases

Paris, France

Applied state-of-the-art time-series prediction models to analyze large-scale EEG data from patients with neurodegenerative diseases.

Research Assistant | Universidad de Buenos Aires

February 2024 - December 2024

EEG-ViT comparison study

Buenos Aires, Argentina

Analyzed EEG data from subjects viewing images, comparing the timing of human visual processing with the activation layers in Vision Transformers (ViT).

NLP Researcher | Universidad de Buenos Aires

June 2023 - December 2023

LLM impact on employment study for Ministry of Labour

Buenos Aires, Argentina

Researched for the Ministry of Labour on the impact of large language models (LLMs) on job roles and potential changes in employment due to technology.

RELEVANT PUBLICATIONS AND CONFERENCES

Academic Publications

“The Variance Brain Foundation Models Forgot: Third-Order Statistics Predict Cognition Where Billion-Parameter Models Fail”

Marraffini, G., Mahuas, G., Borrell, T., Shevchenko, V., Wassermann, D.

Under review

“Do Large Language Models Show Biases in Causal Learning? Insights from Contingency Judgment”

Carro, M., Mester, D., Gauna Selasco, F., Marraffini, G., Leiva, M., Simari, G., Martinez, V.

NeurIPS 2025 (CogInterp workshop)

“Are Optimal Algorithms Still Optimal? Rethinking Sorting in LLM-Based Pairwise Ranking with Batch-ing and Caching”

Wisznia, J., Bolaños, C., Gianolini, A., Hsueh, N., Marraffini, G., Tollo, J., Del Corro, L.

ACL 2025 (Main conference)

“The Greatest Good Benchmark: Measuring LLMs’ Alignment with Utilitarian Moral Dilemmas”

Marraffini, G., Cotton, A., Hsueh, N., Wisznia, J., Fridman, A., Del Corro, L.

EMNLP 2024 (Main conference)

Conference Presentations

OHBM 2026

Bordeaux, France

June 2026

Poster presentation

NeurIPS 2025

San Diego, California

December 2025

2 Accepted papers

ACL 2025

Vienna, Austria

July 2025

Accepted paper main conference

KHIPU 2025

Santiago, Chile

March 2025

Poster presentation

EMNLP 2024

Miami, Florida

November 2024

Accepted paper main conference

SAN 2024

Buenos Aires, Argentina

October 2024

Poster main conference

INDUSTRY EXPERIENCE

AI consultant | Sigma Nova

March 2026 - Present

Pretraining and finetuning fMRI foundation models

Paris, France

Designed and maintained training and evaluation pipelines for large-scale fMRI foundation models on Scaleway clusters. Managed experiment tracking, compute provisioning, and reproducibility tooling across academic and industry environments.

AI Developer | Data Voices

April 2025 - December 2025

AI solutions for conversational responses and document reorganization

Remote

Developed and deployed efficient AI solutions for international companies with requirements ranging from real-time conversational responses to large-scale text-document reorganization.

NLP Engineer | Lumina Labs

December 2023 - May 2025

Multi-agent RAG pipeline for financial presentations

Remote

Developed a multi-agent pipeline for structured RAG in financial presentations. This involved fine-tuning LLMs, enhancing reranking algorithms for retrieval, prompt engineering, and implementing MLOps for deployment on Azure.

TEACHING EXPERIENCE

Lead AI instructor | Anyone AI

March 2026 - Present

Teaching frontier AI to Software Engineers

Remote

Making Software Engineers and coding experts transition into a AI centered careers by teaching most recent advancements and best practices used by the industry.

Teaching Fellow | Universidad de Buenos Aires

June 2024 - February 2025

Algorithms courses for Data Science and Computer Science

Buenos Aires, Argentina

Taught introductory and advanced algorithm courses for Data Science and Computer Science students.

Teaching Assistant | Universidad Torcuato Di Tella

March 2023 - August 2024

Graph Theory and Computational Business Applications

Buenos Aires, Argentina

Taught Graph Theory and Computational Business Applications.

Founder - Paper Reading Club | Universidad de Buenos Aires

September 2023 - December 2024

Academic reading club for data science students

Buenos Aires, Argentina

Founded and led a paper reading club for data science and computer science students to stay updated on the latest research.

EDUCATION

Paris-Saclay University

2026 - Present

PhD in Computational Neuroscience

Paris, France

Universidad de Buenos Aires

2022 - 2024

Master in Data Science

Buenos Aires, Argentina

GPA: 9.30/10

Universidad de Buenos Aires

2019 - 2022

Bachelor of Science in Data Science

Buenos Aires, Argentina

GPA: 8.75/10

FREELANCE PROJECTS

Full Stack Developer

2019 - 2022

Developed web applications using HTML, JavaScript, CSS, Bootstrap, SQL, Python, PHP.

Cryptocurrency Miner

2019 - 2022

Managed cryptocurrency mining operations.

TECHNICAL SKILLS

- **Technical knowledge:** LLM finetuning, Pretraining foundation models, NLP, RAG, Prompt Engineering, Data Analysis, Machine Learning, Artificial Neural Networks, Transformers, Vision Transformers, Statistical Models, Monte Carlo Simulation, Operations Research, Linear Programming, Optimization, Graph Theory.
- **Programming languages:** Python, C++, R, Java, JavaScript, C, Haskell
- **Technologies and frameworks:** Azure, SQL, Langchain, HuggingFace, OpenAI, Pyserini, Langsmith, PyTorch, MNE, Nilearn, fmriprep
- **Languages:** Spanish (Native), English (Proficient), French (low-intermediate)